

Xiaopeng Li

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Professional Experience

The Chinese University of Hong Kong, Shenzhen (CUHK(SZ))
Assistant Professor (tenure track) in the School of Artificial Intelligence.

Shenzhen, China
Sept 2025 – Now

Education

Columbia University
Ph.D. in Operations Research.
M.S. in Operations Research.
Advisor: Cédric Jozs

New York, U.S.
Sept 2020 – May 2025
Sept 2020 – May 2021

The Chinese University of Hong Kong, Shenzhen (CUHK(SZ))
B.S. in Applied Mathematics.

Shenzhen, China
Sept 2016 – May 2020

Research Interests

- Nonconvex and nonsmooth optimization, with applications in large language model reasoning.
- Stochastic optimization for machine learning.
- Applied semialgebraic geometry.
- Dynamical systems.

Preprints & Publications

- Nan, T., Li, X., Kroer, C., & Lin, T. (2026). Efficient Exploration for Iterative Nash Preference Optimization. *arXiv preprint arXiv:2606.01382*. Submitted to NeurIPS 2026.
- Chen, P., Li, X., Chen, X., & Lin, T. (2026). Reward-free Alignment for Conflicting Objectives. *arXiv preprint arXiv:2602.02495*. Accepted by ICML 2026 (Oral, 0.7%).
- Chen, P., Li, X., Li, Z., Chen, X., & Lin, T. (2026). Stepwise Guided Policy Optimization: Coloring your incorrect reasoning in GRPO. *Transactions on Machine Learning Research*.
- Chen, P., Li, X., Li, Z., Yin, W., Chen, X., & Lin, T. (2025). Exploration vs Exploitation: Rethinking RLVR through Clipping, Entropy, and Spurious Reward. *ICLR 2026*.
- Fougereux T., Jozs, C., & Li, X. (2024). *Global convergence of gradient descent for phase retrieval*. *arXiv preprint arXiv:2410.09990*.
- Jozs, C., Lai, L., & Li, X. (2024). Proximal random reshuffling under local Lipschitz continuity. *arXiv preprint arXiv:2408.07182*. Accepted by Mathematics of Operations Research on June 21, 2026.
- Jozs, C., & Li, X. (2024). Singular perturbation in heavy ball dynamics. *Journal of Dynamics and Differential Equations*, 38(1), 233-250.
- Jozs, C., Lai, L., & Li, X. (2023). Convergence of the Momentum Method for Semialgebraic Functions with Locally Lipschitz Gradients. *SIAM Journal on Optimization*, 33(4), 3012–3037.
- Jozs, C., & Li, X. (2023). Certifying the Absence of Spurious Local Minima at Infinity. *SIAM Journal on Optimization*, 33(3), 1416–1439.

Teaching Experience

Instructor

CUHK(SZ)

- AIE1902 AI Exploration II, Spring 2026.
- AIE6003 Optimization for Machine Learning, Spring 2026.

Teaching Assistant

Columbia University

- IEOR E 4007 Optimization Methods & Models for Financial Engineering, Fall 2024
- EEOR E 6616 Convex Optimization, Spring 2024.
- IEOR E 3404 Simulation, Spring 2021.

Undergraduate Student Teaching Fellows

CUHK(SZ)

- Ordinary Differential Equations, Spring 2019 & Spring 2020.
- Elementary Real Analysis, Fall 2019.
- Mathematical Analysis I & II, Fall 2017 & Spring 2018.

Presentations

- Department of Applied Mathematics, The Hong Kong Polytechnic University, invited talk, Hong Kong, June 17th, 2026, *Reward-free Alignment for Conflicting Objectives*.
- INFORMS Optimization Society Conference, invited talk, Atlanta, March 20th, 2026, *Stepwise Guided Policy Optimization: Coloring your incorrect reasoning in GRPO*.
- International Conference on Continuous Optimization, invited talk, University of South California, July 23rd, 2025, *Proximal random reshuffling under local Lipschitz continuity*.
- Young Researcher Workshop, invited talk, Cornell University Ithaca, Oct. 10th, 2024, *Proximal random reshuffling under local Lipschitz continuity*.
- NYC Operations Day, poster, New York, May 10th, 2024, *Convergence of the momentum method for semi-algebraic functions with locally Lipschitz gradients*.
- INFORMS Annual Meeting, invited talk, Phoenix, Oct. 18th, 2023, *Convergence of the momentum method for semi-algebraic functions with locally Lipschitz gradients*.
- Young Researcher Workshop, poster, Cornell University Ithaca, Oct. 1st, 2023, *Convergence of the momentum method for semi-algebraic functions with locally Lipschitz gradients*.
- SIAM Conference on Optimization, invited talk, Seattle, June 2nd, 2023, *Certifying the absence of spurious local minima at infinity*.
- NYC Operations Day, poster, New York, May 5th, 2023, *Certifying the absence of spurious local minima at infinity*.
- INFORMS Annual Meeting, invited talk, Indianapolis, Oct. 18th, 2022, *Certifying the absence of spurious local minima at infinity*.
- International Conference on Continuous Optimization, contributed talk, Lehigh University, July 26th, 2022, *Certifying the absence of spurious local minima at infinity*.

Honors & Awards

- Outstanding Teaching Assistant Award in Industrial Engineering and Operations Research (Columbia University, 2025)
- Academic Performance Scholarship (CUHK(SZ), 2017, 2018 & 2019)
- Dean's List (CUHK(SZ), 2017, 2018 & 2019)
- Undergraduate Research Award (CUHK(SZ), 2018 & 2019)
- Second Prize in China Undergraduate Mathematical Contest in Modeling (China, 2018)
- Second Prize in National College Students Mathematical Competition (China, 2018)

Professional Service

Session Chair

- International Conference on Continuous Optimization, University of South California, July 23rd, 2025.
- International Conference on Continuous Optimization, Lehigh University, July 26th, 2022.

Reviewer

- SIAM Journal on Optimization (Journal)

- Mathematics of Operations Research (Journal)
- Operations Research (Journal)
- Applied Mathematics & Optimization (Journal)
- US Office of Naval Research (Grant)

Mentoring

Undergraduates

- Théodore Fougereux (École Polytechnique). Project: *Bounded flows in phase retrieval*. Summer Internship at Columbia University. May 1 – Sept 31, 2024.
- Cristian Pena (Florida Atlantic University). Project: *Moving Object Detection*. Columbia University Summer Undergraduate Research Experience (SURE) Program. May 29 – July 31, 2024. Underrepresented minority student, Hispanic.

Research Assistant

- Xiaoyan Fan (Master). Project: *Bregman divergence may not preserve boundedness*. Nov 1, 2025 – Sept 30, 2026.
- Xiaogeng Liu (Master). Project: *Global convergence of mirror descent method for functions definable in o-minimal structures*. Feb 1 – Aug 31, 2026.

Internship

National Renewable Energy Laboratory

Golden, CO, USA

Graduate PhD Intern

June 2024 – Aug 2024

- Smoothed progressive hedging algorithm for multi-stage stochastic MILP (available in mpi-sppy).
- L1-penalized ADMM algorithm for solving transmission expansion problems.

Citadel LLC

New York, NY, USA

Quantitative Researcher Intern

June 2025 – Aug 2025

- Reformulated a large-scale nonconvex portfolio optimization problem to exploit structure and improve conditioning, achieving 40% faster solves with KNITRO's barrier (direct) method on 1000 stocks and 30 time periods.
- Established existence guarantees for optimal portfolios in a propagator cross-impact model with factor risk by deriving sufficient, checkable conditions for well-posedness.